LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	46
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of pitch and roll control interceptors
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of pitch and roll control interceptors.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS block diagram and signal flow; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define pitch and roll control interceptors.
(c) How does the AFCS assist the pilot in maintaining aircraft attitude?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• AFCS General Functionality • Pitch Control Interceptor Logic • Roll Control Interceptor Logic • Sensor Inputs to Interceptors • Actuator Response Mechanisms	L & Ds	PPT, IB	15 min
(b)	• Error Signal Generation Process • Gain Adjustment Parameters • Stability and Damping Effects • Manual Override Procedures • Interception Time Constants	L & Ds	PPT, PDF	20 min
(c)	• System Failure Modes • Redundancy in Pitch/Roll Loops • Troubleshooting Interceptor Signals • Impact of External Disturbances • Maintenance Checks for AFCS	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the pitch interceptor corrects attitude errors.
(b) Describe the role of the roll interceptor in lateral stability.
(c) Identify two factors affecting interceptor response time.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from LP 02 to this topic, highlighting how AFCS principles apply to the next lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min